ensemble attack	CW_∞		CW_2		CW_0		avg DSR	exposed model
	most	LL	most	LL	most	LL		
TM	0	0	0	0	0	0	0	TM
DM 1	0	0	0	0	0	0	0	DM 1
DM 2	0	0	0	0	0	0	0	DM 2
DM 4	0	0	0	0	0	0	0	DM 4
DM 5	0	0	0	0	0	0	0	DM 5
black-box	1	0.98	0.99	0.97	0.92	0.91	0.962	TM
grey-rand	0.77	0.85	0.88	0.94	0.74	0.87	0.842	TM, DM 1,2
grey-fix	0.49	0.58	0.66	0.78	0.47	0.65	0.605	TM, DM 1,2
white-rand	0	0	0	0	0	0	0	TM, DM 1-7
white-fix	0	0	0	0	0	0	0	
DistPerturb	0.922	1.189	0.725	0.972	1.614	2.046	-	
DistPercept	21.76	33.78	9.81	18.12	12.58	23.77	-	
Time(s)	350.4	362.5	12.43	17.52	402.8	445.3	-	

TABLE 12: Defense success rate under model verification ensemble CW attack on CIFAR-10.

CIFAR-10 using the target DenseNet model and the 5 verification models listed in Table 3. For the grey box scenarios, we assume that the adversary knows 2 out of the 5 verification models (e.g., VM 1, 2). For the white-box scenarios, we assume that all 5 VMs are known to the adversary. In both cases, ensemble models with a ensemble size of 4 will be chosen. Table 12 shows the results, measured by the defense success rate DSR on each verification model or ensemble option. It also includes the attack costs for the white-box scenario in terms of perturbation distance, perception distance and time cost.

Empirical Analysis. (1) The robustness of our model verification defense approach deteriorates as the adversary increases his knowledge of the composition of the defense system. XEnsemble defense achieves the robustness in terms of average DSR of 84.2% for grey-rand scenario, compared to 60.5% for grey-fix. (2) Our approach fails completely with 0% DSR for white-box scenarios (both white-fix and white-rand). Table 12 also lists the attack costs for the white-box scenario. We see that the cross-model ensemble attack shows the increased perturbation distance, increased perception distance and significantly increased time cost when compared to Table 1. We are encouraged by this preliminary study result for two reasons. First, both grey-box and white-box attacks are only experimented with our output model verification ensemble layer with a small pool of 5 base models on CIFAR-10. We did not incorporate the input-denoising layer ensemble, which could add k times more alternative verifiers and also make the ensemble attack harder to generalize. Second, the output model verification ensemble defense used in this set of experiments did not leverage Kappa diversity, which is another defense parameter, if unknown to the grey-box adversary, would boost the robustness of our defense approach.

Takeaway. we conjecture that (1) extending the size of the baseline candidate set for output-layer verification model ensemble is an important optimization since it can significantly increase the cost for adversaries under grey-box or white-box scenarios; (2) by incrementally adding new models of high test accuracy to expand the baseline candidate model pool over time, we can substantially boost the robustness of XEnsemble defense against grey-box and white-box attacks, turning the white-box attack into grey-box; and (3) by introducing randomization to the input denoising layer, we will have the opportunity to further strengthen the robustness and survivability of our defense approach.

6 Conclusion

We have presented XEnsemble, an input verification and output verification ensemble defense methodology with three novel features. First, XEnsemble is capable of defending a DNN model under protection by utilizing both model-structure diversity and model-disagreement diversity. Second, XEnsemble improves the robustness of the target DNN model by providing auto-repairing and auto-detection capability in the presence of adversarial examples and out-of-distribution examples. Finally, the proposed XEnsemble defense is attack independent. It does not require retraining the target DNN model and can generalize over attack algorithms as well as different out-of-distribution datasets. Evaluated over eleven attack algorithms and two out-of-distribution datasets, we show that XEnsemble can achieve a high defense success rate on both adversarial and OOD inputs.

Our research continues along two directions: (1) We plan to introduce randomization in both input denoising ensemble layer and output model verification layer to add additional robustness to the XEnsemble framework against insider attacks under grey box and white box threat models to defense systems. (2) We plan to extend the XEnsemble approach to other modalities, such as video, audio, and text against adversarial attacks, such as [52].

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